

ICT508 – Information Technology Project Management

Project Scenario Description

BACKGROUND

This project scenario will be used in all assignments and many workshop activities throughout the semester. You will have the opportunity to experience different project roles, such as working as a product owner in an agile environment (Assignment 1) and leading a plan-driven project as a project manager (Assignment 2).

Assume you are working for *Maternity Stream International Pty Ltd* (MSI). MSI is a small to medium-sized enterprise based in Perth, Western Australia (WA), with staff also located in Rockingham and Mandurah. Up to this point, MSI has been focused on health service delivery solutions to WA Health.

There is a growing trend in increased ICT use for the provision of maternity services (Capasso, 2024). WA is well placed from a network and infrastructure perspective to leverage this trend in support of increased population growth from the mining sector. Also, the population in regional WA has increased, resulting in varying and diverse health needs. It is the hope of WA Country Health Service that remote communities, Aboriginal and culturally and Linguistically Diverse (CALD) people, have access to various health services that meet unique and cultural needs. As many stakeholders explore possible solutions, MSI wants to consider a solution that enables virtual healthcare for all people to access urgent care, specialised care, and key maternal services from home, hence reducing the need to travel.

They propose building a SaaS-based healthcare solution. Increasingly, the adoption of Healthcare Software as a Service (Healthcare SaaS) is being viewed by many healthcare entities as an option for delivering maternity services over the Internet. The implementation of a SaaS solution for maternity care can provide expectant mothers with an alternative means of connecting with their treating team. Utilising cloud-based software solutions is considered the most effective way to enhance accessibility to maternity services across various regions in the state. Intended use cases for this solution include:

- Telehealth: enable remote/virtual consultations and remote patient monitoring, expanding access to care and improving patient convenience
- Appointment Scheduling: help healthcare practices manage appointments and other administrative tasks
- Billing management and other claims processing.
- Data analytics: helps healthcare organisations analyse patient data to identify trends, improve care strategies, and optimise operations.
- Health information management and exchange: help labs manage samples, track results, and ensure quality control.

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With this solution, MSI must ensure that SaaS providers have robust security and data protection measures in place. The organisation must ensure that SaaS solutions comply with strict regulations to protect patient privacy.

Given the level of interest in the product, if this project is successful, the company is looking to expand into other Australian markets.

The proposed solution involves the **implementation of a Healthcare SaaS¹ platform – called Maternity Stream System (MSS)** – to support information exchange and the delivery of four critical maternity services, including pathology, imaging, pharmacy, and telehealth. The preliminary intention is to launch a one-year Pilot Project that will establish the foundation for the Maternity Stream System (MSS) and, if successful, expand on this through subsequent project extensions. However, for the moment, the focus will be limited to the *Pilot Project*.

The MSI Chief Information Officer (CIO) (Mr Ben Dover) is the delegated Project Sponsor, and he will be reporting directly to the MSI Chief Executive Officer (CEO) (Mrs Donna Jenner). *As a product owner, in assignment 1, you will work closely with Product Manager Dr Sara Adams, who reports to the CIO.* This means that you are responsible to the product manager and CIO for implementing and overseeing the *MSS software development (sub-Project I)* from initiation to delivery. Later, in Assignment 2, you will assume the role of project manager in implementing the *MSS network infrastructure (Sub-Project II)* and *data migrations (Sub-Project III)* – See project implementation timeline provided.

As a first step, you will need to develop key project initiation and planning materials for the *Board's approval for the Project Charter*. These materials must clearly outline the business case and demonstrate how the project can be successfully implemented (*if approved*). Some of the materials to develop include a project charter and a project management plan – see details in the Assignment 1 Information Sheet.

THE PROJECT

The objectives of the project

The aim of this project is to create a stable, secure, effective, scalable and user-friendly maternity services solution that enhances the provision of four services to expectant mothers – the target services include pathology, imaging, pharmacy and telehealth (remote patient consultations). MSS is a SaaS-based clinical information system² aimed to serve clients in Perth, Rockingham and Mandurah. The system will enable allied health staff to access resources, attend virtual meetings, serve regional patients, and participate in patient-reported outcome measures deployed over the Internet.

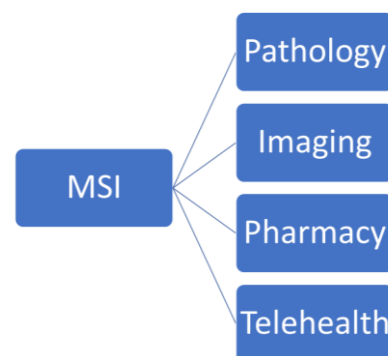


Figure 1. Intended MSS Core Functions

¹ A "Healthcare SaaS" refers to a cloud-based software application specifically designed for the healthcare industry, allowing healthcare providers to access and manage patient data, clinical tools, and administrative functions through an internet connection instead of relying on local servers, typically accessed through a subscription model; essentially, it's a software service tailored to healthcare needs hosted on the cloud.

² A clinical information system (CIS) is a computer-based system used in healthcare settings to collect, store, manage, and readily access patient data like medical history, lab results, imaging studies, medications, and clinical notes, allowing healthcare professionals to make informed decisions at the point of care and streamline workflows by supporting tasks like charting and medication administration; essentially, it's a digital platform that centralises patient information for optimal clinical decision-making.

To support MSI systems, you will need to implement a range of other elements. A preliminary technical architecture has been developed for this system. This architecture and some descriptions of key system elements are available through the LMS under the Assessment Information heading. You will need to ensure that you understand the project deliverables, MSS architecture, and proposed project timeline.

Considerations and regulatory requirements include patient data privacy, security protocols, and identity and access management.

Key issues to be addressed

To implement this project, the following important issues will need to be addressed:

- The steering committee are still discussing whether MSS is a viable and **feasible** solution. The materials that you provide in this Assignment will, therefore, have to be convincing and provide necessary details on viability, feasibility and direction.
- You will need to engage one or more experienced contractors who can coordinate/implement this solution. Identifying and selecting projects and contractors will be a high priority for your efforts.
- You may also want to think about aspects such as leasing infrastructure and other equipment, if needed, to help minimise up-front costs.
- At the end of the project, we will need to ensure that MSI can maintain and manage the maternity services platform it created without the ongoing support of the project team. This will be achieved via a project to support the (P2S) handover process. NB: This is where the SME will provide a handover to the in-house IT support team. This may mean that we need to bring in skills and Intellectual Property (IP) into the company or establish reliable relationships with other organisations that can do this for us.

YOUR BUSINESS EXECUTIVE TEAM

These people/organisations will then be engaged as necessary to support the project.

Person	Roles and additional information
Steering Committee	The steering committee will determine whether and how the project will proceed. This committee is more specifically tasked with (i) ensuring the project is moving within the strategic priorities of MSI, (ii) making fast and high-impact decisions (including changes), (iii) addressing project risks, (iv) supporting stakeholder engagement, and (v) being accountable for project outcomes. Within the board, the CEO, CIO, Program Manager, Product Owner, Subject Matter Experts for critical health services, hospital/clinician representatives, and female representatives from the maternity women's network will be present.
Dr Donna Jenner, PhD (Founder & CEO)	Donna is responsible for the Board and is also the primary Executive Member of the Board. In particular, she is responsible for MSI operations. The CEO will have some strategic oversight requirements for the Project, and you will need to provide some reports to her through the Project Sponsor. However, she would not normally get involved in the day-to-day running of the project; but she would occasionally be involved in, say, quarterly reviews and releases, as well as strategic pivots in the direction of the project. Her major focus is on business outcomes, strategic alignment, risks, and value created. Donna is a standing member of the steering committee for the MSS project.
Ben Dover	Ben is the Chief Information Officer (CIO) for the organisation and is therefore primarily focused on broader aspects than this project. The CIO is therefore

(CIO & Project Sponsor)	responsible to the CEO for all aspects of digital transformations at MSI. As the Project Sponsor, he will be responsible for the strategic coordination of this project and its budget. You and the product manager will need to proactively keep him up to date on progress. Additionally, you would need to engage him in key decisions. However, he would not typically get involved in the day-to-day project work (unless you and the program manager fail to do your job). Ben will be moderately involved, meeting on a bi-weekly or monthly basis, and his primary focus is on technology strategy, resources, and risk management. Ben is also a standing member of the steering committee for the MSS project.
Dr Sara Adam (Product Manager)	Sara is responsible to the CIO and a key member of the steering committee. In particular, she will work with you, the product owner, and stakeholders to define the product strategy and roadmap. The CEO will have some strategic oversight for the Project, and you will need to provide updates to her. However, she would not normally get involved in the day-to-day activities of the project.
Bill Sykes (Legal Support)	He is a legal and contractual expert. Bill can support contract development and negotiations with suppliers and contractors. Additionally, he can assist with the development of license agreements (e.g., End User License Agreements (EULAs)) that will be necessary for client engagement.
Don Morrison (Security & Compliance Expert)	Don is our expert on security issues and standards (e.g., implementing the solutions in accordance with critical standards such as Australian Privacy Principles). He is an outside consultant but will be available as an internal stakeholder to guide security activities once the project commences. Don has experience in healthcare solutions and will help ensure MSS complies with health standards, such as HL7 and FHIR.
Perera Kinley (Data migration expert)	Kinley is our data migration expert with experience in data migration in SaaS implementation. Like Don, he is an outside consultant who was hired and will be available from the project's commencement until its closure.
Ash Mclelland (QA and Test Lead)	Ash will be responsible for ensuring the quality and successful deployment of the SaaS solution by coordinating all testing activities, including test planning, test environment setup, test case design, risk assessment, and facilitating UAT.
Third-Party Providers and subcontractors	Based on the preliminary RFT investigation, you have identified three companies that may be able to meet at least part of the project's needs. These are StreamTech, DCPlus, and DemSet. Information on third-party providers is provided in a separate document on the LMS

YOUR AGILE TEAM COLLABORATIONS

For the **Agile team** implementing the **Maternity Stream System (MSS)**, a mix of technical, clinical, and business roles will be included to ensure the platform meets the needs of users within the healthcare ecosystem. You, the product owner, will work daily with the agile coach (here referred to as the Scrum Master, as the team will use Scrum practices) and the development team.

Your **Agile team** is a cross-functional team of skilled professionals who will collaborate to deliver a high-quality MSS solution incrementally and iteratively. As a team, you will emphasise flexibility, collaboration, and customer-centric development.

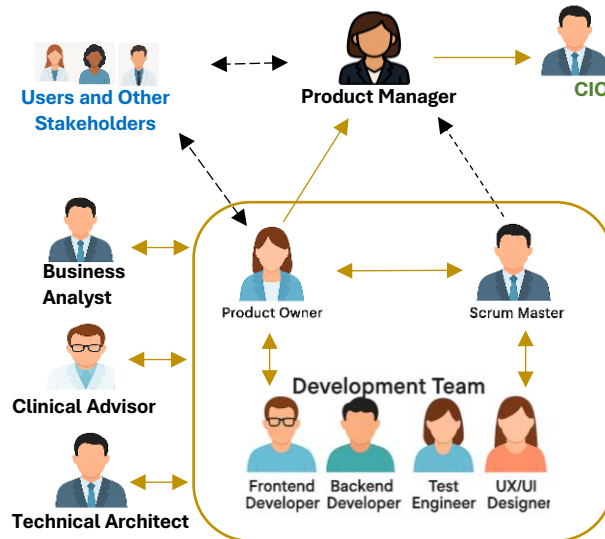


Figure 2. Agile Team Chart

Agile Team for the MSS software	Core responsibilities
You, the Product Owner (PO)	You will represent stakeholders and users, define and prioritise the product backlog, and ensure the MSS delivers value to users and meets organisational needs. You report to the Product Manager, Dr Sara Adam.
Agile Coach (<i>known as Scrum Master in Scrum</i>)	An agile coach will facilitate agile ceremonies, remove distractions & blockers, and ensure the team follows Agile principles. They will set up the project teamwork environment so that the team can work efficiently.
Development Team	You will have a cross-functional Dev Team including: • Frontend Developers to build user interfaces for clinicians, patients, and admins. • Backend Developers to develop APIs, data exchange mechanisms, and system logic. • DevOps Engineers to manage deployment pipelines, cloud infrastructure, and system reliability. • QA/Test Engineers to ensure system quality through automated and manual testing. • UX/UI Designers to ensure intuitive interfaces are created and tailored to healthcare workflows.
Business Analyst (BA)	A BA with experience in digital health solutions to collaborate with PO to refine user stories, acceptance criteria, business process modelling, impact analysis, and translate clinical and business needs into data and technical requirements.
Clinical Advisor	The clinical advisor is a subject matter expert (SME) who will advise on healthcare workflows
Technical Architect	The technical architect is an expert in technical design, who will help to ensure that the solution is scalable, secure, and maintainable. This person will assist in choosing appropriate frameworks, standards, and identify technical risks

To build the MSS right, you will need to work with or consult with subject matter experts (SMEs). Clinical Advisor (a Midwife or Obstetrician) to provide guidance on functions, validate workflows and terminology. A data scientist will be consulted on analytics and reporting features. A telehealth specialist will be consulted on remote care workflows for the telehealth. The Pharmacy Integration lead will be consulted on medication management features.



Figure 3. MSS Users

User involvement is critical to project success. Besides users who also double as domain experts, like midwives, clinicians, and telehealth providers. The MSS will have dedicated user representatives, including a hospital administrator representative, to provide input on scheduling, billing, and operational workflows. Additionally, patient advocates will provide input on patient-related needs, including the accessibility and user-friendliness of patient-facing features.

Information about Users is provided on the LMS.

PRELIMINARY COST/REVENUE INFORMATION

A preliminary analysis has indicated that the following costs and revenues are likely to accrue from the implementation of the Project. These are Rough Order of Magnitude (ROM) figures only, but they should provide a useful starting position for the analysis.

Description	Year 0	Year 1	Year 2	Year 3
ROM Costs/Expenditures	\$705,000	\$590,000	\$420,000	\$360,000
ROM Benefits/Revenues	\$256,000	\$815,000	\$1,120,000	\$1,331,000

Assumptions

The following base assumptions need to be taken into account when analysing the preceding ROM cost/revenue figures and the associated project aspects:

- These figures do not take into account the discount effects related to inflation and the cost of capital.
- The inflation rate is expected to remain at about 3% per annum for the next four years.
- This project will be funded through debt (*a loan from the Bank*). There will be no establishment costs (*as this will be an extension of existing funding agreements*), and the bank has offered a fixed loan rate of 4.8% per annum.
- Year 0 commences at the Project Start Date. The following years are consecutive.
- The ROM expenditures take into account the software licensing fees for the SaaS platform, costs associated with consulting on configuration, system integration, and data migration provided by the SaaS vendor and third-party providers.
- Software licensing is expected to take 38% of the budget, implementation services will take 25%, user training and support 15%, data migration and cleansing 10%, loan servicing/financing costs may take 5%, and contingency allowance 10%.

- Revenues in Year 0 relate to project feasibility study, project planning, stakeholder consultations and engagement of clients after User Acceptance Tests (UAT).
- As these are ROM estimates, variances of 25-40% may be expected

ANTICIPATED PRODUCT RELEASE

Most milestones are related to the release plan, in which you should be able to release a set of usable features of the MSS every quarter (every 3 months). It is anticipated that the most critical functions of the MSS software will all be finalised within 12 Months.

OVERALL PROJECT KEY MILESTONES

The following key milestones are crucial to the overall success of the MSS Project. Please note that these only reflect some of the high-level objectives that are pertinent. You will need to work out other key milestones involved. The subordinate milestones that you select must allow the following to be achieved.

Date	Event/Milestone
01 Mar 2026	• Preparations Commence. You will begin developing informal team relationships (e.g., start talking to potential contractors and internal stakeholders) and complete the development of the materials required by the Board (the project charter) and other documents as requested in Assignment 1.
13 Apr 2026	• Submission to the Board. Submit the required project materials to the Board so they can decide whether to proceed with the project.
25 May 2026	• Board Approval (Stage Gate Approval for the Project Charter/Business Case). The project charter and other documents submitted to the board are approved.
01 Jun 2026	• Planned formal project Start Date. This milestone will trigger the formal allocation of project funds from this date, and it will only be initiated if the Board approves your submission. Until this is authorised, other staff and providers cannot be engaged formally on a paid basis.
01 Jun 2026	• Engage Providers. A formal RFT process will be conducted to select and contract a Healthcare SaaS provider, technical implementation contractors, and relevant third-party specialists. Evaluation criteria will include healthcare experience, regulatory compliance, technical capability, security maturity, cost, and ability to provide ongoing Level 2/3 support. This milestone ensures that appropriately qualified providers are engaged prior to entering the execution phase and reduces delivery and operational risk.
15 Jun 2026	• Commence MSS Software Development (Sub-Project I). Formal commencement of agile software development activities for the Maternity Stream System (MSS) following provider engagement and project approval.
01 Dec 2026	• Stage Gate Approval for the Sub-Project II and III Execution phase. The MSI staff and engaged providers will need to complete the solution designs and developments prior to this date; after this, the project will enter the execution phase – see this stage gate on the project timeline.
26 Mar 2027	• Network and Infrastructure Installation and Commissioning. Networks and Infrastructure work in Subproject II has been ongoing, and by this date, the required IT infrastructure and network environment for the MSS are installed, configured, tested and validated.
05 May 2027	• Go-Live Approval. This is the date that all services are completed and go live. The Board have identified this as a hard-wall milestone (e.g. it cannot be moved). Before this date, a full one-month awareness campaign will be conducted for local healthcare providers and referrers. • To avoid any possible embarrassment after opening, the entire system must be fully operational and stable by this date. • From an infrastructure perspective, all three sites are identical (<i>i.e., Perth, Rockingham and Mandurah</i>)
27 Aug 2027	• Formal Pilot Project Closure. The Pilot Project will be closed (See Topic 11), and final reports and other deliverables must be provided by this date. If the Pilot is successful, the project is intended to continue, but the Pilot Project Closure process must still be done by this date.

When engaging providers, MSI will undertake the following processes.

1 Tender Preparation (We prepare the documents that contain questions and key information, which defines what we want from other organisations.)	• Project definition and scoping (identify our requirements and the priorities that will be used in the weighted analysis); • Identify appropriate tenderers (so we can go out to market with the RFT); • Develop the tender documentation and draft contracts; and • Send out the RFT, so appropriate organisations can respond.
2 Tender Period (Other organisations answer our questions by explaining their experience, what they can do to meet our needs, and what it would cost.)	• Responding organisations answer the questions/points in the RFT (this includes explaining their expertise and providing specific solutions to the needs identified in the RFT); • Responding organisations will develop pricing for delivering their proposed solutions; • During the tender period the organisation that prepared the tender may answer questions from responding organisations and provide amendments as necessary (to clarify points); and • Prior to the closing date/time the responding organisations will provide their RFT responses.
3 Tender Evaluation (Our team evaluates the tender responses that are received and determines the most appropriate organisation/s to deliver our needs.)	• The responses are typically short listed to ensure that there are only a few analysed in detail (this is greatly simplified if you only sent the tender to a few contractors); • A detailed analysis will be used to determine a total score for proposed deliverables; • Each tenderer's total score will be assessed in relation to cost, to determine a Value for Money outcome; and • The organisation/s that score best in terms of the Value for Money outcomes will be nominated as Preferred Tenderer (one or more as required).
4 Contract Negotiations & Signature (From the previous step, one or more Preferred Tenderer/s will be selected. We will then do final negotiations with them and once all parties have reached an agreement the contract/s will be signed.)	• A detailed negotiation will be conducted, which allows the organisation developing the tender and Preferred Tenderer/s to reach a common understanding and agreement on the issues related to the tendered services/systems; and • Once the agreement is reached, the contractor and the organisation developing the tender will sign the contract and this initiates the formal agreement.

FURTHER INFORMATION ABOUT THIS PROJECT AND UNIT ASSIGNMENTS

You can now proceed to read the Proposed System Technical Architecture, Users, and the proposed Project Timeline. These documents are provided on the LMS in the section 'Project Scenario Information'. Once you have a good understanding of the project proceed to Assignment 1 Information Sheet